AGGREGATE SHOCKS AND DEBT MANAGEMENT II

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LAST TIME

- Barro (1979) – incomplete markets, reduced form:
  - Government minimizes deadweight loss of taxes;
  - Only non-state-contingent, risk-free debt;
  - Result: tax smoothing $\Rightarrow$ taxes are a random walk, debt absorbs shocks.
- Lucas and Stokey (1983) – complete markets, general equilibrium:
  - Full set of Arrow securities;
  - Primal approach: a single implementability constraint;
  - Result: allocations and taxes depend only on the current state s_t, not on history.
- Very different predictions for the behavior of taxes and debt!

THIS LECTURE

- Aiyagari, Marcet, Sargent, and Seppälä (2002) – AMSS:
  - Same Lucas–Stokey environment (preferences, technology, Ramsey problem);
  - But the government can only issue risk-free one-period debt;
  - General equilibrium with incomplete markets.
- Main question: what happens to optimal taxes and debt when state-contingent claims are not available?
- Answer: a blend of Barro and Lucas–Stokey.

THE AMSS MODEL

ENVIRONMENT

- Same environment as in part I, with the same history notation:
  - s_t is a finite-state Markov shock and government purchases are $g(s_t)$;
  - Resource constraint: $c(s^t) + g(s_t) = A\,\ell(s^t)$;
  - Preferences: $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u\left(c(s^t), \ell(s^t)\right)$;
  - Flat-rate labor income tax $\tau(s^t)$.
- Key difference: the government can only trade one-period risk-free debt $b_{t+1}(s^t)$.
  - $b_{t+1}(s^t)$ is chosen at t after observing s^t;
  - it promises the same payoff for every successor state s_{t+1}.

GOVERNMENT BUDGET CONSTRAINT

- Under risk-free debt, the government budget constraint is:

$$g\left(s_t\right) + b_t\left(s^{t-1}\right) = \tau\left(s^t\right) A\,\ell\left(s^t\right) + p_t^b\left(s^t\right)\, b_{t+1}\left(s^t\right)$$

  where $p_t^b\left(s^t\right)$ is the price of a one-period risk-free bond.

- Same timing convention as in part I:
  - $b_t\left(s^{t-1}\right)$ is the payoff due today, chosen at $t-1$;
  - $b_{t+1}\left(s^t\right)$ is the new promise chosen today.
- From the household Euler equation,

$$p_t^b\left(s^t\right) = \sum_{s_{t+1}} q\left(s^{t+1} \mid s^t\right) = \beta\, \mathbb{E}_t\left(\frac{u_c\left(s^{t+1}\right)}{u_c\left(s^t\right)}\right).$$

- Debt is bounded: $\underline{M} \leq b_{t+1}\left(s^t\right) \leq \overline{M}$.

IMPLEMENTABILITY CONSTRAINTS

COMPLETE MARKETS: A SINGLE CONSTRAINT

- Recall from Lucas–Stokey: competitive equilibrium allocations satisfy a single implementability constraint:

$$\sum_{t=0}^{\infty} \sum_{s^t} \beta^t \pi\left(s^t\right)\left[u_c\left(s^t\right) c\left(s^t\right) + u_\ell\left(s^t\right) \ell\left(s^t\right)\right] = u_c\left(s^0\right) a_0$$

- One multiplier Φ — constant across time and states.
- This is equivalent to the time-0 present value government budget constraint.

GOVERNMENT SURPLUS IN TERMS OF ALLOCATIONS

- Recall from Lucas–Stokey: using the household intratemporal FOC, $\left(1-\tau\left(s^t\right)\right)A = -u_\ell\left(s^t\right)/u_c\left(s^t\right)$, we can write the government surplus entirely in terms of allocations:

$$z\left(s^t\right) \equiv \tau\left(s^t\right)A\,\ell\left(s^t\right) - g\left(s_t\right) = c\left(s^t\right) + \frac{u_\ell\left(s^t\right)}{u_c\left(s^t\right)}\ell\left(s^t\right)$$

- $z\left(s^t\right)$ is the net-of-interest surplus: tax revenues minus government purchases.
- It depends on the allocation $\left(c\left(s^t\right), \ell\left(s^t\right)\right)$ and on exogenous $g\left(s_t\right)$.
- No prices or tax rates appear — this is the primal approach.

PRESENT VALUE OF FUTURE SURPLUSES

- Define the present value of government surpluses from date t onward:

$$B_t\left(s^t\right) \equiv \mathbb{E}_t\left(\sum_{j=0}^{\infty} \beta^j \frac{u_c\left(s^{t+j}\right)}{u_c\left(s^t\right)} z\left(s^{t+j}\right)\right)$$

- The ratio $\beta^j u_c\left(s^{t+j}\right)/u_c\left(s^t\right)$ is the household's stochastic discount factor from the Euler equation — it converts date-$(t+j)$ surplus into date-t consumption units.
- This is derived from the household's FOCs, not from assuming Arrow–Debreu markets exist.
- $B_t\left(s^t\right)$ is the amount the government can repay at date t given its future fiscal plans.
- In any competitive equilibrium: $B_t\left(s^t\right) = b_t\left(s^{t-1}\right)$ (the government debt maturing at t).

THE MEASURABILITY CONSTRAINT

- Under complete markets: $b\left(s^t\right)$ is an Arrow security payoff — it can depend on the current state s_t.
  - Different B_t for different realizations of $s_t \Rightarrow$ no extra restriction.
- Under incomplete markets: $b_t\left(s^{t-1}\right)$ is risk-free — it cannot depend on s_t.
  - The government promised to repay b_t last period, before knowing what s_t would be.
  - Equivalently, for every $t \geq 1$ and predecessor history s^{t-1}, there exists a number $b_t\left(s^{t-1}\right)$ such that

    $$B_t\left(s^{t-1}, s_t\right) = b_t\left(s^{t-1}\right)$$

    for every current state s_t.
- In a finite-state model, "measurable w.r.t. s^{t-1}" is exactly this: for a fixed predecessor history, the date-t value cannot vary across current states.

RAMSEY CONSTRAINTS UNDER INCOMPLETE MARKETS

- This is a restriction on allocations: (c, ℓ) must be such that the discounted value of future surpluses is known one period ahead.
- Plus debt limits:

$$\underline{M} \leq b_t\left(s^{t-1}\right) \leq \overline{M}.$$

- These constraints replace the flexibility that state-contingent Arrow securities provided.
- Summary of constraints on the Ramsey allocation:
  1. Implementability constraint (eq. 8) — same as Lucas–Stokey;
  2. Measurability: $B_t\left(s^t\right)$ is measurable w.r.t. s^{t-1} (eq. 10) — new;
  3. Debt limits (eq. 9) — new.

WHY DO MEASURABILITY CONSTRAINTS MATTER?

- Consider a positive g shock. Under complete markets:
  - The surplus z_t falls (more spending, possibly lower tax revenue);
  - But $B_t\left(s^t\right)$ can fall too — the government had bought "war insurance" (Arrow claims that pay in bad states).
- Under incomplete markets:
  - B_t is fixed regardless of s_t — no insurance payoff;
  - The government must adjust future allocations to generate higher future surpluses: raise taxes, cut spending, or borrow more;
  - This is the source of history dependence and a strong unit-root component in taxes and debt.
- Incomplete markets force the government to spread the cost of a shock over the entire future — rather than absorbing it through a state-contingent portfolio.

THE RAMSEY PROBLEM WITH INCOMPLETE MARKETS

LAGRANGIAN

- Attach multipliers to the constraints and use iterated expectations. The Lagrangian is (AMSS eq. 11):

$$L = \mathbb{E}_0\left(\sum_{t=0}^{\infty} \beta^t \left[u\left(c_t, \ell_t\right) - \psi_t u_{c,t} z_t + u_{c,t}\left(\nu_{1t}\overline{M} - \nu_{2t}\underline{M} + \gamma_t b_{t-1}^g\right)\right]\right)$$

- From here on I follow the paper's compact time-subscript notation: $g_t \equiv g\left(s_t\right)$, $c_t \equiv c\left(s^t\right)$, $\ell_t \equiv \ell\left(s^t\right)$, $b_t^g \equiv b_t\left(s^{t-1}\right)$. The recursive state is $\left(s_t,\ b_{t-1}^g,\ \psi_{t-1}\right)$.
- The key multiplier ψ_t evolves as:

$$\psi_t = \psi_{t-1} + \nu_{1t} - \nu_{2t} + \gamma_t$$

  with $\psi_{-1} = 0$.
- γ_t is the multiplier on the measurability constraint at t.
- $\nu_{1t}, \nu_{2t} \geq 0$ are multipliers on the debt limits.

COMPLETE VS. INCOMPLETE MARKETS: ROLE OF ψ_t

- Under complete markets: $\gamma_{t+1} = \nu_{1t} = \nu_{2t} = 0$ for all $t \geq 0$, and γ_0 is the (scalar) multiplier on the present-value budget constraint.
  - $\psi_t = \psi_0 = \gamma_0$ — constant across time and states.
- Under incomplete markets: ψ_t changes each period because $\gamma_t \neq 0$ in general.
  - ψ_t is a new state variable — it summarizes the accumulated effect of past shocks on the shadow cost of the government budget.

WHAT THE MULTIPLIERS MEAN

- Do not confuse three different objects:
  - λ in part I: the household multiplier on its present-value budget constraint, i.e. the marginal utility value of one extra unit of date-0 wealth;
  - Φ in Lucas–Stokey: the Ramsey multiplier on the implementability constraint;
  - AMSS write w_t; here I call the same object ψ_t.
- ψ_t is the continuation analogue of Φ: it measures the welfare effect of tightening the government's continuation financing problem from date t onward.
- γ_t is the multiplier on the date-t measurability/news constraint, so off the debt limits:

$$\psi_t - \psi_{t-1} = \gamma_t.$$

- In the paper's sign convention, $w_t \leq 0$: more required present-value government spending lowers welfare. Up to sign, it is the shadow cost of public funds.

THE MULTIPLIER ψ_t IS A RISK-ADJUSTED MARTINGALE

- In the paper, the first-order condition with respect to b_t^g gives (AMSS eq. 16):

$$\mathbb{E}_t\left(u_{c,t+1}\,\gamma_{t+1}\right) = 0.$$

- If debt limits do not bind at $t+1$, then $\nu_{1,t+1} = \nu_{2,t+1} = 0$, so

$$\psi_{t+1} = \psi_t + \gamma_{t+1}.$$

- Substituting $\gamma_{t+1} = \psi_{t+1} - \psi_t$ into eq. 16 gives (AMSS eq. 17):

$$\psi_t = \left(\mathbb{E}_t\left(u_{c,t+1}\right)\right)^{-1}\,\mathbb{E}_t\left(u_{c,t+1}\,\psi_{t+1}\right).$$

- So ψ_t is a risk-adjusted martingale: it is a martingale under marginal-utility weights, not under the physical measure unless $u_{c,t+1}$ is constant.

WHERE ψ_t ENTERS THE PLANNER'S FOCS

- In the paper's compact notation, the date-t FOC with respect to c_t is:

$$u_{c,t} - u_{x,t} - \psi_t k_t + \left(u_{cc,t} - u_{cx,t}\right)\left(\nu_{1t}\overline{M} - \nu_{2t}\underline{M} + \gamma_t b^g_{t-1}\right) = 0$$

  where $x_t = 1 - \ell_t$ and k_t is the same curvature term as in Lucas–Stokey (AMSS eq. 14).
- Under complete markets, this collapses to

$$u_{c,t} - u_{x,t} - \gamma_0 k_t = 0,$$

  with one constant multiplier γ_0 (AMSS eq. 15).
- Under incomplete markets, ψ_t moves with history and the inherited debt term b^g_{t-1} also enters through the constraint multipliers.
- This is the cleanest way to see why today's taxes depend on current g_t and on history.

FROM ψ_t TO TAXES

- The Ramsey first-order conditions imply that taxes and allocations at date t depend on:
  1. s_t — the current shock, through $g(s_t)$ (as in Lucas–Stokey);
  2. b^g_{t-1} — the inherited debt level;
  3. ψ_{t-1} — the accumulated multiplier (a near–unit root process).
- State variables: $\left(s_t,\ b^g_{t-1},\ \psi_{t-1}\right)$.
- The dependence on ψ_{t-1} introduces history dependence — unlike Lucas–Stokey, but like Barro.

SPECIAL CASE: QUASI-LINEAR PREFERENCES

QUASI-LINEAR UTILITY ELIMINATES PRICE MANIPULATION

- AMSS's example 2 normalizes $A = 1$ and is easiest to write with labor:

$$u(c_t, \ell_t) = c_t + H(1 - \ell_t),$$

  where H is increasing and strictly concave.
- Then $u_c = 1$ always, so:
  - Bond price is constant: $p^b = \beta$;
  - Government cannot manipulate intertemporal prices.
- The Ramsey problem reduces to a consumption-smoothing problem:

$$\max_{\{R_t, b_t^g\}} \mathbb{E}_0 \left(\sum_{t=0}^{\infty} \beta^t W(R_t) \right) \qquad \text{s.t.} \quad b_t^g \geq \beta^{-1} \left(g_t + b_{t-1}^g - R_t \right)$$

  where R_t is tax revenue and $W(R)$ is the Barro-style return function induced by the household problem.

TWO CASES: NATURAL VS. AD HOC ASSET LIMITS

- Natural asset limit: government assets bounded by $\underline{M} = -g_{\max}/(1-\beta)$.
  - ψ_t is a submartingale bounded above by zero $\Rightarrow$ converges a.s. to zero;
  - $\tau_t \to 0$: taxes converge to zero;
  - Government accumulates a war chest of assets $g_{\max}/(1-\beta)$, financing all spending from interest.
- Ad hoc asset limit: $\underline{M} > -g_{\max}/(1-\beta)$.
  - Occasionally the asset limit binds;
  - ψ_t does not converge — it continues to fluctuate as a martingale away from the boundaries;
  - Taxes and debt exhibit Barro-like random walk behavior.

COMPLETE MARKETS: CONSTANT TAX RATES

- For comparison, under complete markets with quasi-linear utility:
  - the planner can insure the fiscal shock perfectly;
  - tax rates are constant across time and states.
- This is the extreme of “tax smoothing across states” — the government uses state-contingent debt as insurance.
- Under incomplete markets, even with the same preferences, taxes fluctuate and exhibit a strong near–unit-root component.

GENERAL PREFERENCES

CAN INCOMPLETE MARKETS REPLICATE COMPLETE MARKETS?

- With absorbing states for g: yes — once g stops changing, ψ_t converges to a constant and the allocation coincides with a Lucas–Stokey allocation (for a different initial debt level).
- With persistent randomness in g and curvature in u: no.
- **Lemma 3** (AMSS): if the primary deficit under the Lucas–Stokey solution is sufficiently random, then the incomplete markets allocation cannot converge to any complete markets allocation.
  - Intuition: the complete markets surplus is random, so under risk-free debt, equation $b_{t+1} = r_t (b_t - s_t)$ would make debt explosive.
- So the incomplete markets economy must distort the Lucas–Stokey surplus process enough to prevent explosive debt dynamics and keep debt within its bounds.

SUMMARY OF ASYMPTOTIC POSSIBILITIES

Case	Debt limits	g process	Outcome
Example 2	Natural	Stationary, $u_c = 1$	$\tau_t \to 0$ (war chest)
Example 2	Ad hoc	Stationary, $u_c = 1$	Barro-like fluctuations within bounds
Example 3	Natural	Absorbing state	Converges to Lucas–Stokey tail allocation
General	Natural	Stationary, u curved	Does not converge

- The last two rows summarize the paper's Section IV/V results, which are stated under natural debt and asset limits.
- In the general case, ψ_t has a nondegenerate limiting behavior rather than settling down to the Lucas–Stokey constant-multiplier case.
- Tax rates blend Lucas–Stokey (response to current g) with Barro (persistent drift from ψ_t).

NUMERICAL EXAMPLES

PARAMETERIZATION

- AMSS Section VI: CRRA utility, rewritten in our labor notation:

$$u(c,\ell) = \frac{c^{1-\sigma_1} - 1}{1-\sigma_1} + \eta \frac{(1-\ell)^{1-\sigma_2} - 1}{1-\sigma_2}$$

  with $(\beta, \sigma_1, \sigma_2, \eta) = (0.95, 0.5, 2, 1)$.
- The paper's first numerical exercise uses i.i.d. continuous purchases with mean $\bar{g} = 30$ and scale factor $\alpha = 0.4$.
- For the theory, nothing essential changes if we replace that process by a finite-state Markov approximation.
- Debt limits: $(\underline{M}, \overline{M}) = (-1000, 1000)$.
- Initial debt: $b^g_{-1} = 0$.

TAX RATE AUTOREGRESSIONS

	Complete	Incomplete
$\mathbb{E}[\tau]$	0.3108	0.2776
$\mathrm{sd}(\tau)$	0.0018	0.0191
AR(1) coefficient b	-0.0054	0.9888
R^2	2.9×10^{-5}	0.9944

- Under incomplete markets: AR(1) coefficient $\approx 1 \Rightarrow$ near–unit root in taxes.
- Taxes are lower on average but much more volatile.

PAPER SIMULATIONS: WAR FINANCE

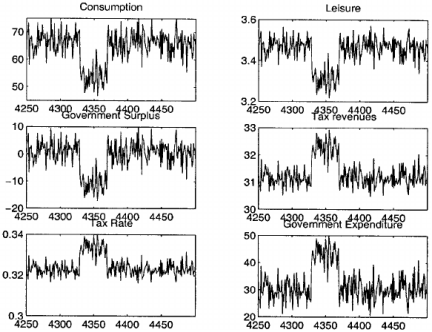

FIG. 6.—Simulation of peace and war economy with complete markets

War Finance under Complete and Incomplete Markets

We computed another example with regime-switching government ex-

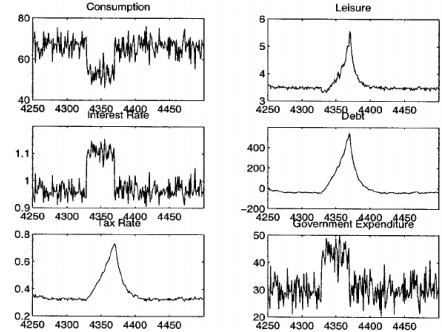

FIG. 7.—Simulation of peace and war economy with incomplete markets

omies with identical parameter values and government expenditure pro-

AMSS Figures 6 and 7. The paper plots leisure, so $x_t = 1 - \ell_t$ in our notation. Incomplete markets show wartime borrowing and gradual postwar repayment.

KEY TAKEAWAYS

SUMMARY

1. Restricting the government to risk-free debt retains the Lucas–Stokey implementability constraint but adds sequences of measurability constraints.
2. The multiplier ψ_t on these constraints is a risk-adjusted martingale, introducing a unit root component into taxes and debt.
3. Taxes depend on current state s_t (like Lucas–Stokey) and on history through ψ_{t-1} and b^g_{t-1} (like Barro).
4. With quasi-linear utility and natural debt limits, the government accumulates a war chest and taxes $\to 0$. Ad hoc asset limits restore Barro's random walk.
5. In the general case with persistent randomness and utility curvature, the allocation cannot converge to complete markets.

LUCAS AND STOKEY'S THREE LESSONS, REVISITED

1. Budget balance in present value must be respected.

   $\Rightarrow$ Supported.

2. No case for continual budget balance.

   $\Rightarrow$ Amplified: debt is a buffer stock that absorbs fiscal shocks.

3. State-contingent debt is important.

   $\Rightarrow$ Qualified: depends on persistence and variance of g.

   For i.i.d. shocks, self-insurance via risk-free debt nearly replicates complete markets.

   For persistent war shocks, welfare loss is significant.